

1) List of Companies

→ Global Tech & product Companies

Google

Microsoft

Uber

Amazon

Adobe

Oracle

Atlassian

Intuit

Samsung

→ Fin Tech & Indian startups

Flipkart

Phonepay

Mynted

Swiggy

Razerpay

Walmart

↳ IT Services & Consulting

Infosys

Deloitte

Wipro

IBM

Cognizant

ACC

2) Syllabus of DSA in Gate Exam:-

i) programming & Data structures

↳ programming in C: Basic concepts, functions, recursion & parameter passing

↳ Data structures: Arrays, stacks, queues, linked lists, trees, binary search trees (BST), binary heaps, and graphs

↳ Algorithm Analysis: Asymptotic Notations

(O , Ω , Θ) and worst / average case time & space complexity analysis

↳ Searching and Sorting;

Linear Search, Binary Search, Sorting Algorithms
(Merge sort, quick sort, Selection sort, bubble sort
and insertion sort).

↳ Algorithm Design Paradigms;

Greedy Algorithms, divide and Conquer,
and dynamic programming

↳ Graph Algorithms;

Traversals (breadth-First search and Depth-
First search); Minimum spanning

(Kruskal's & prim's) & shortest path Algorithms

(Dijkstra's & Bellman - Ford)

↳ Hashing & Basic Complexity;

Hashing Techniques & an Introduction to NP
Completeness

3) Variables in Amazon

invoice - id (string)

order - id (string)

invoice - date (int)

Customer - name (string)

Address (string)

email (string)

Invoice mail

items (List)

product - name (Str)

Quantity (int)

Total (Double)

tax (Double)

shipping - fee (Double)

each-item price (double)

Current Bill

bill id → string
Customer id → string
meter num → (string)
Reading date → (int)
Volume used → double
net amount → double
govt subsidy amount → double
payment status → Boolean

Gas Bill

Customer name → string
Consumer no → unsigned long int
Billing Date → int
Due Date → int
Total Amount → Double
LPD ID → unsigned long int
Address → string
Mobile No → unsigned long int